

SOC Homelab Lab Report

Wazuh SIEM — Phase 1 & 2

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What was done

What happened

What it means

Extracted Report Content

This report documents the setup and operation of a personal SOC (Security Operations Centre) homelab built using Wazuh, an open-source SIEM and XDR platform. The lab was constructed across two phases: Phase 1 covered infrastructure setup and agent enrolment, and Phase 2 covered attack simulation and alert detection mapped to the MITRE ATT&CK; framework.

The goal of this project is to demonstrate hands-on capability in endpoint monitoring, threat detection, and security event analysis — skills directly applicable to SOC analyst and blue team roles.

The lab consists of two physical devices connected on the same local network:

Firewall (UFW) rules configured on the manager:

The Wazuh agent (v4.14.5, Apple Silicon build) was downloaded, installed, and configured on the MacBook to communicate with the Ubuntu manager.

Key steps performed:

After a successful start, the following five agent processes were confirmed running:

Agent macbook-pvee (ID 001) appeared as Active in the Wazuh dashboard. The MacBook was now a fully monitored endpoint, forwarding authentication events, file integrity data, and system logs to the Ubuntu manager in real time.

With the agent enrolled and the manager collecting telemetry, Phase 2 involved simulating real-world attack techniques against the Ubuntu manager from the MacBook. The goal was to verify that Wazuh correctly detects, categorises, and maps these events to the MITRE ATT&CK; framework.

An SSH brute force loop was executed from the MacBook terminal, repeatedly attempting to authenticate as a non-existent user (wronguser) against the Ubuntu manager:

The Ubuntu machine received repeated SSH authentication attempts from 10.135.235.5. Each attempt was rejected with an invalid user error. The auth.log on the Ubuntu machine recorded entries including:

This simulates a credential stuffing or brute force attack (MITRE ATT&CK; T1110), where an attacker systematically attempts SSH logins with incorrect credentials. Wazuh's log collection module picked up the auth.log entries and fired the corresponding alerts automatically. The volume and frequency of failures triggered both individual failure rules and the higher-severity repeated failure pattern rule.

The following Wazuh rules triggered during the simulation and were visible in the Threat Hunting → Events view:

Total events recorded across the 24-hour window: 436 hits, including 35 authentication failures and 40 authentication successes (the latter from legitimate sudo operations on the manager itself).

Wazuh's built-in MITRE ATT&CK; module automatically mapped the detected events to adversary tactics and techniques. The following were observed in the MITRE ATT&CK; dashboard view:

The following views were confirmed active and populated in the Wazuh dashboard at the time of the simulation:

This lab demonstrates the following capabilities relevant to a SOC analyst or blue team role:

The following items are planned to extend the lab further:

Key Bullet Points

- Dashboard served over HTTPS on port 443, using a self-signed TLS certificate
- Listening on 0.0.0.0:443, accessible from any device on the local network

- HTTP 200 responses confirmed in server logs from client IP 10.135.235.5
- Downloaded the arm64 agent package from the official Wazuh package repository
- Installed using macOS installer with administrator privileges
- Edited `/Library/Ossec/etc/ossec.conf` to set the manager address to 10.135.235.2
- Started all agent processes using `wazuh-control restart`
- Agent registered automatically with the manager on port 1515 and was assigned ID 001, name `macbook-pvee`
- Enrol the Ubuntu machine itself as a second agent to monitor the manager host
- Add Suricata as a network IDS, feeding network-layer events into Wazuh
- Integrate the VirusTotal API for automated file hash reputation checks
- Write a custom Wazuh decoder and rule to detect a specific log pattern
- Simulate additional MITRE techniques: T1033 (System Owner Discovery), T1565 (File Integrity Modification)
- Document the full lab on GitHub as a portfolio project with architecture diagram and screenshots

Commands & Log Entries

```
for i in {1..10}; do ssh -o StrictHostKeyChecking=no
wronguser@10.135.235.2 2>/dev/null; done
```

```
Failed password for invalid user wronguser from 10.135.235.5 port 57154
ssh2
```

```
Invalid user wronguser from 10.135.235.5 port 57158
```

```
syslog: User missed the password more than one time
```